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Stacked Polyphase Bridge Converters for High-Voltage Traction Drives: A Review and System-Level Design Perspectives

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ABSTRACT Stacked polyphase bridge (SPB) converters partition a high dc-link voltage among series-connected three-phase bridges, each supplying an electrically separated machine winding group. This architecture enables lower-voltage semiconductor devices and modular converter–machine integration, but requires coordinated modulation, voltage balancing, and fault management. This paper reviews SPB topology evolution, operating principles, machine integration, semiconductor scaling, and control strategies for high-voltage traction drives. It synthesizes the effects of carrier interleaving on dc-link ripple, the conditions and limitations of common-mode cancellation, and the balancing behavior of different control structures. The review connects cell count and device voltage class with winding configuration, gate-drive complexity, cooling, and post-fault operation. These interdependencies inform an iterative system-design framework, illustrated by a high-voltage traction example. The synthesis shows that semiconductor-level benefits must be evaluated alongside the additional converter and machine requirements, and that the preferred cell count depends on the application and operating mission. Joint battery–converter–machine design are identified as priorities for establishing the practical competitiveness of SPB traction drives.

INDEX TERMS Electric vehicles, integrated motor drives, modular converters, multiphase machines, stacked polyphase bridge (SPB) converters.

I. Introduction

Electrification is pushing traction-drive propulsion and fast-charging power toward several hundred kilowatts; meanwhile, system-level performance requirements continue to rise [1]. At that point, increasing the dc-link voltage has emerged as a primary design lever for meeting both demands. On the propulsion side, a higher dc-link voltage can improve efficiency and power density [2]. On the battery and charging side, it reduces the current required for a given power level, lowering cable conduction loss and conductor cross-section while enabling higher charging power under a given current limit [3].

This trend, however, is not without consequence at the converter level. A higher dc-link voltage also brings more severe insulation-coordination and electromagnetic-compatibility (EMC) demands, together with added thermal and reliability stress on the power stage [2], [3]. Raising the dc-link voltage therefore requires coordinated converter, machine, and thermal design.

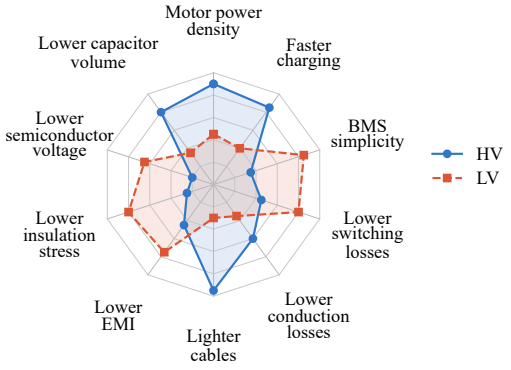


FIGURE 1. Conceptual illustration of qualitative system-level tradeoffs between low-voltage and high-voltage traction architectures.

Fig. 1 summarizes these qualitative tradeoffs between conventional low-voltage and higher-voltage traction architectures.

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TABLE 1. Representative voltage and power levels in published traction-inverter studies, industrial and reference platforms, and production battery-electric vehicles.

Published studies / demonstrators	Industrial / reference platforms	Production vehicles
Bertelshofer et al.: 400 V, 150 kW [6]	**Bosch Gen. 4**:400/800 V, 250/350 kW peak [7]	**BMW i4 M50 (2022)**: 400 V class, 400 kW max. [8]
Taha et al.: 800 V, 100 kW [9]	**TI/Wolfspeed reference design**: 800 V, 300 kW [10]	**Hyundai IONIQ 5 N (MY2026)**: 800 V class, 478 kW (10-s boost) [11]
Al-Hmoud et al.: 1 kV, 200 kW [12]	**Infineon eCAV platform**: 800 V, 250 kW [13]	**Volvo EX90 Performance (MY2026)**: 800 V class, 500 kW max. [14]
Wang et al.: 1.2 kV, 100 kW [15]	**AVL PI850e**: 800 V, 250 kW [16]	**Porsche Taycan Turbo S**: 800 V class, 700 kW (Launch Control) [17]
Ayaz et al.: 1.2 kV, 250 kW [18]	**Punch IV5**: 800 V, 600 kW [19]	**Lucid Air Dream Performance**: 924 V architecture, 828 kW max. [20]

Vehicle powers are combined motor-output ratings, including boost where indicated, and are not single-inverter continuous ratings. Voltage classes do not equal nominal battery voltage (BMW: 398.5 V; IONIQ 5 N: 697 V). Industrial values follow the respective platform specifications.

A. Literature and Industry Trends Toward Higher DC-Link Voltage

The trend motivated above is not merely conceptual: it is already established in academic prototypes, industrial inverter platforms, and production vehicles, as summarized in Table 1. Reported academic demonstrators range from 400-V to 800-V and 1.2-kV converters, with still higher levels investigated for high-power passenger and heavy-duty applications [3], while industrial platforms increasingly target 800-V operation as a baseline and production vehicles show a comparable shift from established 400-V architectures to a growing number of 800-V and higher-voltage platforms [4], [5]. This shift makes semiconductor blocking-voltage capability a central converter-design constraint.

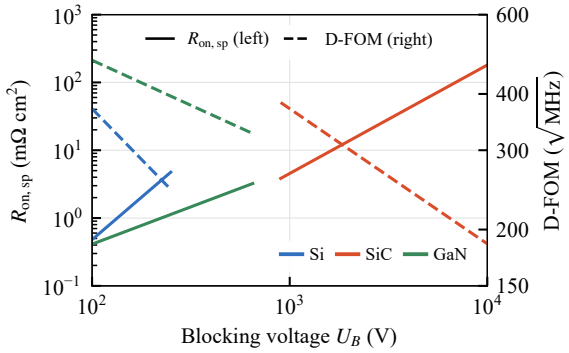


FIGURE 2. Specific on-state resistance and D-FOM versus blocking voltage, using empirical device fits from [23].

B. Semiconductor Blocking-Voltage Tradeoff

In a conventional two-level voltage-source inverter, each switch must withstand the dc-link voltage. Increasing the blocking-voltage rating of a unipolar device generally increases its specific on-state resistance, linking system voltage to conduction loss [21], [22]. Switching performance also depends on output charge. Following [23], these effects are represented by

$$R_{\mathrm{on,sp}} = k_R U_B^{\alpha_R}, \qquad C_{\mathrm{oss,Q,sp}} = k_C U_B^{\alpha_C},$$

where U_B is the device blocking voltage and the empirical coefficients are taken from [23]. The resulting device figure-of-merit is

$$\mathrm{D{-}FOM} = \frac{1}{\sqrt{R_{\mathrm{on,sp}} C_{\mathrm{oss,Q,sp}}}} \propto U_B^{-(\alpha_R+\alpha_C)/2}.$$

Here, $C_{\mathrm{oss,Q}}(U) = Q_{\mathrm{oss}}(U)/U$ is the charge-equivalent capacitance at the evaluated switching voltage. The resistance–capacitance product is independent of die area; a higher D-FOM indicates lower minimum hard-switching semiconductor losses under the model assumptions.

Fig. 2 shows the fitted trends over the approximate voltage ranges of the source survey, with on-state resistance at 25 °C. Lower blocking voltage improves D-FOM within each fitted technology, but comparisons across technologies require consistent voltage and operating conditions. Wide-bandgap devices expand the available design choices [21], [24]–[26]; their system benefit remains dependent on device selection, switching conditions, and converter implementation. Voltage partitioning therefore offers a route to exploiting lower-voltage devices, subject to the additional hardware and control requirements of the topology.

C. Multilevel Converters and Their Limitations

Multilevel converters provide a way to partition voltage and utilize lower-voltage devices. NPC, T-type, flying-capacitor, and cascaded architectures all distribute the dc-link voltage among additional semiconductor positions or capacitor-voltage levels, reducing individual device voltage stress [27]–[30].

They require additional semiconductor positions, passive components, and gate drivers, which increases the number of potential failure points. Classical three-phase multilevel bridges still supply a common winding set, whereas cascaded and other modular architectures can provide module-level redundancy. In either case, fault tolerance depends on available isolation paths, spare voltage capability, and control; component count alone establishes neither fault tolerance nor system reliability.

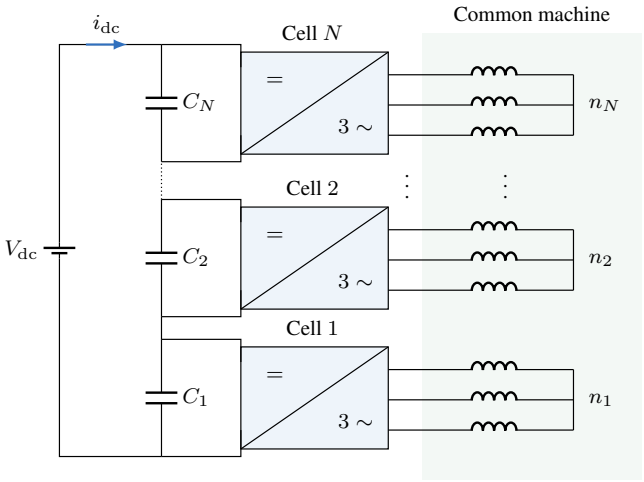


FIGURE 3. Conceptual architecture of an N-cell SPB traction drive: series-connected local dc-link capacitors, series-connected three-phase two-level bridge cells, and their electrically separated machine winding groups.

D. Stacked Polyphase Bridge

The stacked polyphase bridge (SPB) distributes the high-voltage conversion function among N repeated converter units. Each unit is a conventional three-phase two-level bridge connected in series on the dc side, supplying its own electrically separated three-phase winding group of the same machine, as illustrated in Fig. 3.

Under balanced operation, each cell's local dc-link voltage $V_{\mathrm{dc},k} \approx V_{\mathrm{dc}}/N$, so the semiconductor voltage rating can be selected on the local cell voltage rather than the complete traction dc-link voltage – addressing the semiconductor bottleneck much as multilevel converters do. Each cell also drives a separated winding group, providing modularity at the converter–machine interface. A faulted cell can, in principle, be isolated and bypassed while the remaining cells continue to drive the machine at reduced power. It is this combination of semiconductor voltage reduction and converter–machine modularity that distinguishes SPB from conventional multilevel and cascaded architectures and motivates its consideration as a candidate topology for next-generation traction drives.

Realizing this potential, however, introduces coupled design challenges in component count, dc-link dynamics, machine design, common-mode voltage, and fault handling, addressed throughout this review. These challenges have grown more tractable as neighboring research areas have matured: multiphase machines provide additional post-fault degrees of freedom, integrated modular motor drives show how repeated power-electronic units can be physically associated with individual machine sections, wide-bandgap semiconductors amplify performance differences among voltage classes, and distributed control coordinates modular converter units [31]–[36].

Despite more than a decade of SPB research, beginning with Norrga *et al.* [37], the feasibility of electric-vehicle traction applications requires these developments to be assessed together. The available literature remains distributed across converter topology, modulation, dc-link stability, multiphase-machine design, integrated motor drives, semiconductor technology, thermal management, distributed control, and fault-tolerant operation. Although these topics are strongly coupled in an SPB traction drive, they are rarely evaluated within a common system boundary, leaving a gap between a well-motivated architectural concept and a system-level feasibility assessment. Accordingly, this paper provides a system-level review of SPB traction-drive technology aimed at closing that gap.

E. Review Scope and Contribution

This is a focused narrative review rather than an exhaustive systematic survey or a meta-analysis. Direct SPB evidence concerns series-connected dc ports supplying electrically separated winding groups; related multiphase and integrated-drive studies provide enabling methods but are not treated as demonstrations of SPB traction performance. The review distinguishes reported results, analytical illustrations, and application-level inferences. Its contribution is to connect device voltage partitioning to winding feasibility, modulation, balancing, common-mode excitation, and post-fault operation within one EV design boundary.

The remainder of the paper is organized as follows. First, SPB's historical development, operating principle, and modulation are consolidated. Second, the interaction between cell count and the electric machine is examined alongside multiphase-machine and integrated-modular-drive research. Third, common-mode voltage and dc-link dynamics and balancing are examined. Fourth, semiconductor voltage scaling and fault handling are treated as coupled architecture-level design problems. Finally, the remaining research gaps and automotive validation requirements are identified.

II. Evolution of the Stacked Polyphase Bridge Concept

SPB belongs to the broader development of modular converter–machine systems. Modular and integrated modular motor drives established repeated converter–winding units, machine segmentation, power sharing, and fault-tolerant operation [31], [38]. Series-connected implementations extended these concepts to dc-voltage sharing, enabling lower-voltage semiconductor devices [32], [39], [40]. Other modular drive topologies adopted the same principle [41], [42].

Norrga *et al.* introduced SPB in 2013 [37], followed by dedicated control and modulation methods [43]. Related series-connected drives demonstrated internal dc-link regulation using machine-side degrees of freedom [44]. Subsequent stability analyses established the balancing-controller requirements [45], [46], identifying dc-link balancing as a central control challenge alongside semiconductor voltage selection.

Research then expanded toward coordinated modulation, scalable control, and converter–machine design. Modulation

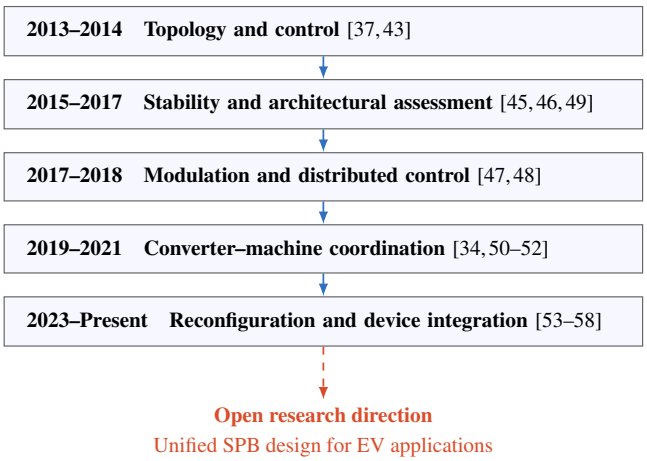


FIGURE 4. Representative developments in SPB research grouped by period and research focus.

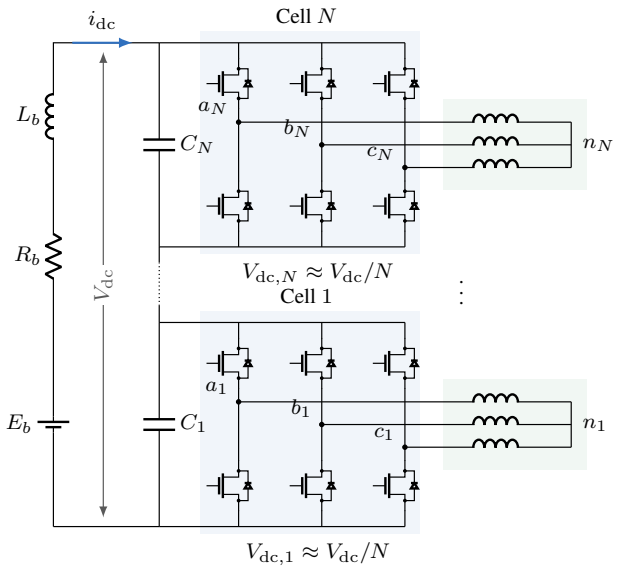


FIGURE 5. Switch-level realization of the SPB dc–ac stage for two representative cells (Cell 1 and Cell N; cells $2, \ldots, N-1$ elided).

studies linked switching loss to dc-link harmonics [47], while distributed control provided an alternative to centralized balancing [48]. Comparative assessment against parallel-connected polyphase bridges and other modular converters evaluated machine design, losses, capacitor energy, cost, and redundancy, positioning SPB among several candidate architectures [49]. Interleaving and phase-count studies further connected converter performance to machine architecture [34], [50]–[52].

Recent work has extended SPB toward decentralized and reconfigurable operation. Common-duty-ratio control, decentralized balancing, and multi-agent power distribution reduce reliance on centralized coordination [53]–[56]. Online module isolation and reintroduction enable dynamic dc-bus reconfiguration, including removal of a faulted cell without interrupting torque production [57]. In parallel, gallium nitride devices have renewed interest in the benefits of lower semiconductor voltage classes [58].

Fig. 4 groups these developments by research focus, tracing the progression from voltage partitioning to coordinated converter–machine design and reconfigurable operation. The remaining integration gap for electric vehicles is a unified, application-specific SPB design framework that couples battery and dc-link requirements, semiconductor selection, machine configuration, and control, evaluated against common energy, power-density, cost, and reliability targets.

III. Operating Principle and Topological Characteristics

A. Basic Structure and Voltage Partitioning

An N-cell SPB consists of N conventional three-phase two-level voltage-source inverter cells connected in series on the dc side, with each bridge supplying one electrically separated three-phase winding group of the same machine [37], [43], [47]. The machine therefore presents N ac ports, while the converter cells share a single series dc path.

The dc-link voltage satisfies

$$V_{\mathrm{dc}} = \sum_{k=1}^{N} V_{\mathrm{dc},k}, \qquad V_{\mathrm{dc},k} \approx \frac{V_{\mathrm{dc}}}{N} \tag{1}$$

under balanced operation. A basic implementation contains $6N$ active switches and N local dc-link capacitors. Increasing N reduces the nominal cell voltage, and hence the required semiconductor voltage class, but simultaneously increases the number of switches, gate drivers, sensors, capacitors, and machine-winding terminals that must be realized and managed. Fig. 5 shows this structure at switch level for two representative cells, making explicit the $6N$-switch bridge realization, the local capacitors C_k, and the notation used throughout the remainder of this paper.

B. Series DC Connection and Power Flow

Neglecting auxiliary and bypass paths, the cell dc currents are constrained to be equal,

$$i_{\mathrm{dc},1} = i_{\mathrm{dc},2} = \cdots = i_{\mathrm{dc},N} = i_{\mathrm{dc}}, \tag{2}$$

so that the dc-side power entering cell k is

$$p_{\mathrm{dc},k} = V_{\mathrm{dc},k} i_{\mathrm{dc}}. \tag{3}$$

The total machine power, by contrast, is the sum of the powers delivered to the individual winding groups,

$$P_{\mathrm{ac}} = \sum_{k=1}^{N} P_k. \tag{4}$$

Under symmetrical operation $P_k \approx P_{\mathrm{ac}}/N$, but equal power sharing is not a fundamental requirement of the topology. The multi-three-phase machine provides additional current and power degrees of freedom that allow power to be redistributed among winding groups [44], [55]. This

redistribution capability is the physical basis for both dc-link voltage balancing and several of the post-fault operating strategies.

The common dc current and independently controlled ac powers make local capacitor voltages internal dynamic states. If one winding group processes more power than another, the corresponding capacitor energies can diverge even though all cells carry the same dc current [45], [46]. Managing this divergence is the dc-link voltage-balancing problem, SPB's first-order control challenge.

C. Cell Count as a Converter–Machine Design Variable

Cell count is consequently a converter–machine design variable rather than a semiconductor variable alone. Increasing N also reduces the voltage available to each winding group and alters the number of winding sets, terminal count, electromagnetic coupling, current-control degrees of freedom, and available reconfiguration states.

Machine phase current, however, does not generally scale as $1/N$: it instead depends on the number of turns, flux linkage, winding connection, torque constant, and machine geometry. A many-cell architecture may therefore reduce the semiconductor voltage class while still requiring high-current devices or parallel chips at the device level.

IV. Electric-Machine Requirements and Integrated Implementation

A. Winding Architecture and Electromagnetic Coupling

Although the winding groups are electrically separated at their terminals, they share the same magnetic circuit, rotor, air gap, and shaft. Their electromagnetic coupling depends on winding distribution, slot/pole combination, spatial displacement, and machine geometry [34], [44], [55], [59]. Therefore, an SPB-compatible machine cannot generally be obtained simply by dividing a conventional three-phase winding into multiple groups without considering the resulting coupling and harmonic behavior.

The three-phase groups may be aligned, spatially shifted, or implemented as physically segmented stator modules. Spatial displacement affects mutual coupling, torque-ripple components, and the additional current subspaces available to the controller, among other effects [60]–[62]. These additional subspaces can introduce non-torque-producing currents, but they can also be exploited for harmonic suppression, differential power sharing, and fault-tolerant operation [55], [63]. Split winding architectures show a similar tradeoff: they can reduce ripple and provide additional current/voltage degrees of freedom, but they do so by increasing winding terminals and coupling complexity [64]. These three winding-group arrangements are compared schematically in Fig. 6.

For the kth PMSM winding group, a steady-state approximation neglecting mutual coupling between groups gives

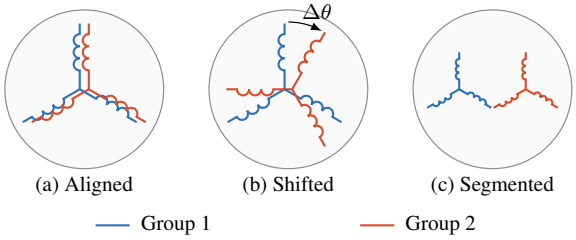


FIGURE 6. Schematic comparison of winding-group spatial arrangements: (a) aligned groups (parallel phase axes), (b) spatially shifted groups (offset by $\Delta\theta$), and (c) physically segmented stator modules.

$$\begin{aligned} v_{d,k} &= R_{s,k} i_{d,k} - \omega_{\mathrm{e}} L_{q,k} i_{q,k}, \\ v_{q,k} &= R_{s,k} i_{q,k} + \omega_{\mathrm{e}} \left(L_{d,k} i_{d,k} + \psi_{f,k} \right), \end{aligned} \tag{5}$$

For coupled or transient operation, the flux-linkage model must retain $\dot{\psi}_{d,k}$ and $\dot{\psi}_{q,k}$ and the dependence of flux on all winding-group currents. The required voltage must satisfy

$$\sqrt{v_{d,k}^2 + v_{q,k}^2} \leq k_{\mathrm{mod}} V_{\mathrm{dc},k} \approx k_{\mathrm{mod}} \frac{V_{\mathrm{dc}}}{N}. \tag{6}$$

This falling per-cell voltage budget ties winding turns, inductance, and field-weakening capability directly to N – a coupling demonstrated in optimized multiphase SPB-PMSM systems [34]. Recent multiphase traction-machine studies similarly emphasize coordinated design of winding voltage, harmonic subspaces, and modulation [65], [66].

B. From Multiphase Machine to Integrated Modular Drive

Physical integration can convert the added electrical complexity of many winding groups into a packaging advantage. With a remote inverter, additional winding groups increase phase-cable and connector count; positioning each converter bridge close to its corresponding winding group instead reduces external cabling, shortens high-frequency interconnections, distributes heat sources, and aligns converter modules with stator segments.

This design philosophy predates SPB and has been extensively studied in integrated and modular motor drives [31], [32], [35], [39], [40], [67]–[77], [79], summarized in Table 2.

C. Physical Integration Options

The electrically separated winding groups required by SPB do not impose a unique converter placement on a radial-flux machine. Chowdhury *et al.* [80], consistent with the broader integrated-motor-drive survey of Abebe *et al.* [81], classify motor/inverter physical integration into four categories: whether the converter mounts radially or axially, and whether it interfaces with a separate housing/end-cap structure or directly with the stator core. This taxonomy frames how SPB converter cells can be placed around a radial-flux machine, as illustrated in Fig. 7. The axial-flux stator-mounted and polygon-housing alternative is covered by the Mohamed *et al.* developments [75]–[77] in Table 2.

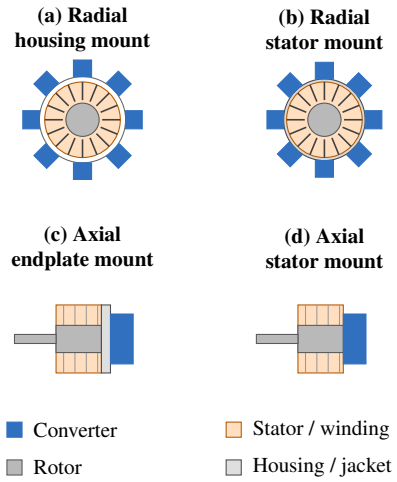


FIGURE 7. Physical integration options for a radial-flux machine, following [80].

TABLE 2. Representative Integrated Motor-Drive Developments

Reference	Integration theme
Shakweh *et al.* [67]	Integrated motor/inverter architecture
Wheeler/Pickering *et al.* [68], [69]	Converter and thermal design inside motor envelope
Brown *et al.* [70]	Converter mounted at modular stator pole
Wolmarans *et al.* [71]	Fault-tolerant converter, machine, and thermal design
Choi *et al.* [31]	Modular PM machine and high-temperature electronics
Wang *et al.* [39], [40]	Low-voltage GaN, dc-voltage sharing, and interleaving
Uğur and Keysan [73], [74]	Multi-physics converter–machine optimization
Mohamed *et al.* [75]–[77]	Stator-mounted modules, polygon housing, integrated dc link
Wang *et al.* [32]	Repeated machine and converter submodules
Chen *et al.* [79]	WBG converter, machine, passive, and cooling co-design
Deng *et al.* [35]	Recent IMD technologies and research directions

D. Integration Is Not Free of Complexity

Integration does not remove complexity; it relocates it. Converter modules mounted on or near the machine are exposed to elevated temperature, vibration, and sealing/coolant requirements, among other stresses, and to electromagnetic coupling between gate-drive, power, and machine conductors. Shared machine–converter cooling can also increase thermal interaction between the stator and power electronics [75].

SPB introduces an additional insulation challenge because its converter cells and associated winding groups occupy different absolute potentials along the series dc stack. Gate-driver supplies and sensing interfaces must preserve electrical isolation between the local power stage and the control system, as illustrated by isolated traction-inverter implementations [82]. Mechanical mounting and thermal interfaces must likewise preserve the required insulation distances and solid-insulation barriers [83].

TABLE 3. SPB modulation degrees of freedom grouped by principal system-level effect.

Effect	Degrees of freedom
Ripple and loss shaping	Carrier phase [47], [52], [84]
	Switching sequence [85], [86]
	Clamping and DPWM [87], [88]
CMV and balancing authority	Zero-sequence injection and vector redundancy [89], [90]
	Differential cell commands [55], [91]

V. Modulation Strategies

Each stacked polyphase bridge (SPB) cell supports conventional sinusoidal pulsewidth modulation (SPWM), space-vector modulation (SVM), and discontinuous pulsewidth modulation (DPWM). However, the shared dc current, dynamic cell voltages, and electromagnetic coupling between winding groups require coordination across cells. Carrier phase, zero-sequence injection, switching sequence, and clamping intervals provide additional degrees of freedom for shaping dc-link ripple, machine harmonics, switching loss, common-mode voltage (CMV), and winding-group power distribution [43], [47]. Table 3 groups these degrees of freedom by their principal system-level effects.

A. Carrier Coordination and Harmonic Cancellation

For a binary upper-switch command $s_{x,k} \in \{0,1\}$, the double-Fourier representation is

$$s_{x,k}(t) = \sum_{m,n\in\mathbb{Z}} c_{mn,x,k} e^{j[m(\omega_c t+\theta_{c,k})+n(\omega_0 t+\theta_{0,k})]}. \quad (7)$$

Here m and n are carrier and fundamental indices. The coefficients include the dc term, baseband harmonics, carrier harmonics, and sidebands; conjugate symmetry gives a real switching function. A common carrier frequency and fundamental frequency are assumed across cells.

The carrier phase is a free parameter for shaping aggregate harmonics. A general carrier assignment is

$$\theta_{c,k} = \theta_{c,1} + \delta_k, \qquad \delta_1 = 0. \quad (8)$$

Uniform interleaving uses $\delta_k = 2\pi(k-1)/N$; Fig. 8 illustrates this case for $N = 3$.

B. DC-Link Current Ripple and Battery-Side Current

DC-link current ripple affects capacitor loading and the current presented to the battery and input filter. Higher switching frequency can reduce passive-component requirements within a specified loss budget, while interleaving shapes the aggregate spectrum.

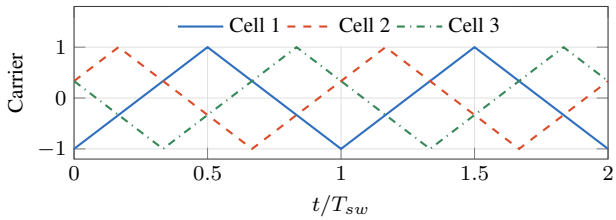


FIGURE 8. **Uniform carrier interleaving for $N = 3$, with phase offsets 0, $2\pi/3$, and $4\pi/3$. Analytically generated triangular carriers.**

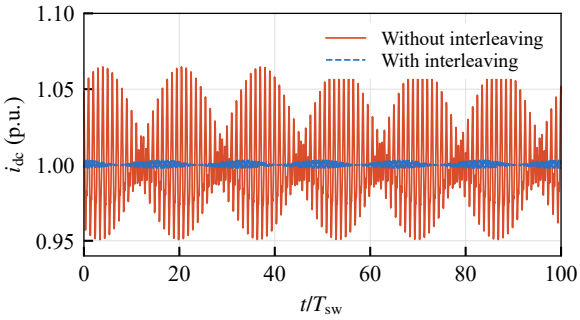


FIGURE 9. **Analytical steady-state shared dc-link current for three identical SPB cells with ideal natural SPWM over one fundamental period. Parameters are $M = 0.85$, power-factor angle $20°$, $f_{sw}/f_1 = 100$, $L_bC/T_{sw}^2 = 0.85$, and $R_bC/T_{sw} = 2.5$.**

With phase current positive from converter to machine, the ideal cell input current is $i_{\mathrm{conv},k} = \sum_x s_{x,k} i_{x,k}$. Consequently, its spectrum is

$$I_{\mathrm{conv},k} = \sum_{x\in\{a,b,c\}} S_{x,k} \otimes I_{x,k}. \tag{9}$$

For matched cell spectra with a common fundamental phase, uniform interleaving multiplies carrier-order m by

$$\sum_{k=1}^{N} e^{jm2\pi(k-1)/N} = \begin{cases} N, & m \in N\mathbb{Z}, \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

This selection rule cancels nonmultiples of N; it does not guarantee a nonzero component exactly at Nf_{sw}. The first surviving term also depends on the switching coefficients, sideband index, and winding/current phase relationships [43]. Unequal cell loading or voltage sharing weakens cancellation.

The shared dc-link current follows from the equivalent circuit in Fig. 5:

$$\begin{aligned} L_b \frac{di_{\mathrm{dc}}}{dt} &= E_b - R_b i_{\mathrm{dc}} - \sum_k v_{\mathrm{conv},k}, \\ C_k \frac{dv_{\mathrm{conv},k}}{dt} &= i_{\mathrm{dc}} - i_{\mathrm{conv},k}. \end{aligned} \tag{11}$$

For equal cell capacitances, $C_k = C$, the relationship between the summed converter-current spectrum and dc-bus current is [43]

$$I_{\mathrm{dc}}(\omega) = \frac{\sum_{k=1}^{N} I_{\mathrm{conv},k}(\omega)}{N - \omega^2 L_b C + j\omega C R_b}. \tag{12}$$

Equation (12) assumes identical capacitors, a fixed source voltage, and prescribed converter-current spectra. For positive-frequency peak phasors, a consistent ripple measure is

$$I_{\mathrm{dc,rms}}^{\mathrm{ripple}} = \left[\frac{1}{2} \sum_{\omega_h > 0} \frac{|\sum_k I_{\mathrm{conv},k}(\omega_h)|^2}{|N - \omega_h^2 L_b C + j\omega_h C R_b|^2} \right]^{1/2}. \tag{13}$$

Contributions at the same frequency are combined before taking the magnitude. Retained harmonics must be evaluated against the input-network resonance; interleaving does not ensure attenuation at every frequency. Fig. 9 illustrates one operating point.

The analytical example includes lumped battery-to-stack inductance but assumes identical cells, prescribed sinusoidal phase currents, ideal switching, and ideal cell connections. In distributed implementations, bus-bar and connector-loop inductances alter harmonic cancellation. For instance, a series-connected gallium nitride integrated modular motor drive required careful capacitor selection and compact layout to preserve interleaving benefits [40]; an earlier characterization found that parasitic loop inductance could make interleaved ripple exceed the noninterleaved value [92].

C. Machine Harmonics, Voltage Utilization, and Switching Loss

The preferred carrier displacement depends on cell count, modulation index, power factor, winding displacement, and the optimization objective. SPB studies demonstrate its influence on semiconductor losses and dc-link current harmonics [47], while related modular and multiphase drives demonstrate simultaneous reductions in dc-link and machine current ripple [52], [85], [93].

In multiphase machines, nonfundamental voltages can excite low-impedance harmonic subspaces, increasing copper loss and torque ripple despite satisfactory fundamental current control. Interleaved and discontinuous modulation strategies therefore require adaptation to the multiphase current-ripple characteristics [84], [94], [95].

At high speed, voltage utilization becomes critical because each winding group is supplied by its local cell voltage $V_{\mathrm{dc},k}$, subject to the machine voltage limit in (6). This constraint becomes more restrictive as cell count increases and local voltage decreases. Optimized vector selection in dual-three-phase traction machines can extend operation in the high-modulation region while suppressing unwanted subspace currents [65].

Switching-loss reduction introduces a competing objective. DPWM and hybrid strategies reduce transitions by clamping selected phases or redistributing high-frequency switching among modules. Related open-end-winding studies demonstrate simultaneous reductions in switching loss and zero-sequence current [87], [88]. Although their dc architecture differs, the principle of coordinating switching activity among inverter modules is applicable to SPB.

D. Coupling to Voltage Balancing and Common-Mode Control

Modulation also determines the control authority available for dc-link voltage balancing and CMV reduction. Voltage balancing requires differential winding-group power commands that remain feasible within local voltage and current limits [55], [56], [91]. Redundant voltage vectors and coordinated zero-sequence injection can reduce common-mode excitation of the machine frame [89], [90], [96]. These objectives are examined in Sections VII and VI, respectively.

SPB modulation consequently requires coordinated switching decisions that account for ripple, losses, voltage utilization, balancing, and common-mode behavior across the operating map.

VI. Common-Mode Voltage Effects

In any converter-fed ac machine, the non-zero average of the three phase-leg pole voltages produced by PWM switching gives rise to a common-mode voltage (CMV) that excites parasitic capacitive paths between winding, frame, shaft, and bearings. This mechanism is well characterized for a converter with a single machine port whose voltage scaling is produced internally, such as the flying-capacitor multilevel converter, for which an analytical CM-EMI model referenced to one dc-link potential exists [97]. SPB does not fit this picture: because its N cells are connected in series on the dc side while each feeds an electrically separate winding group, cell k's local dc rails are displaced from the overall dc-bus reference by an amount that depends on its position in the stack.

A. Series-Stack Origin of Common-Mode Excitation

Taking cell 1's negative dc rail as the reference, with cell k's negative rail coinciding with cell $(k-1)$'s positive rail, the absolute offset of cell k is

$$V_{-,k} = \sum_{j=1}^{k-1} V_{\mathrm{dc},j} \approx (k-1)\frac{V_{\mathrm{dc}}}{N}, \tag{14}$$

using the voltage-partitioning relation (1). The common-mode potential seen by winding group k is then

$$v_{\mathrm{cm},k}^{\mathrm{abs}}(t) = V_{-,k} + \frac{1}{3}\left(v_{aN_k} + v_{bN_k} + v_{cN_k}\right), \tag{15}$$

where $v_{aN_k}, v_{bN_k}, v_{cN_k}$ are cell k's pole voltages referenced to its own negative rail N_k. The second term is the conventional switching-induced CMV already addressed by zero-sequence injection and vector-redundancy modulation; the first term, $V_{-,k}$, is specific to SPB, set by the cell's stack position and by the instantaneous voltage-sharing state, not by modulation alone.

Fig. 10 illustrates the resulting structure: increasing absolute potential offset with stack position, together with representative CM current paths coupling each winding group to the shared frame and chassis ground through parasitic capacitance. Because the cells and their winding groups are physically close together in the integrated implementations, these paths are short and the coupling between cells is direct rather than mediated by long cabling. Physically integrated motor-drive assemblies show that common-mode current can resonate through exactly this kind of parasitic loop, and that the resonance is not adequately predicted by a single-source CM model [96], [98]. Applying these models to SPB requires accounting for multiple machine ports, distinct dc offsets, and their coupled return paths.

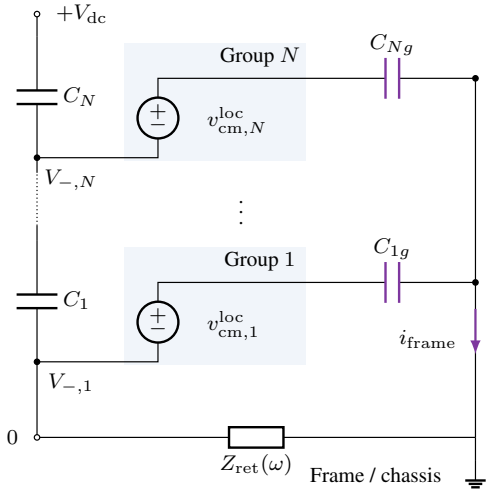


FIGURE 10. Lumped common-mode equivalent of the SPB stack.

B. Carrier Synchronization for CMV Cancellation

Beyond the stack-offset mechanism above, synchronizing each cell's PWM waveform relative to the others is a way to actively cancel the switching-induced component of CMV. We illustrate the principle for the $N = 2$ case, where the cancellation condition has a simple closed form; the same synchronization idea extends to larger stacks built from matched cell pairs, subject to matching and the required winding arrangement.

For equal cell voltages V_c, define the midpoint-referenced term $u_{\mathrm{cm},k} = V_c(\sum_x s_{x,k}/3 - 1/2)$. Complementary commands $s_{x,2} = 1 - s_{x,1}$ give $u_{\mathrm{cm},2} = -u_{\mathrm{cm},1}$; reversing the second winding group's sense preserves the intended torque direction. The absolute voltages remain offset:

$$\frac{v_{\mathrm{cm},1}^{\mathrm{abs}} + v_{\mathrm{cm},2}^{\mathrm{abs}}}{2} = V_c = \frac{V_{\mathrm{dc}}}{2}. \tag{16}$$

Thus switching ripple cancels in the arithmetic mean, not the dc potential relative to the bottom rail. This balanced-cancellation principle was first demonstrated for a dual-winding-set machine driven by a balanced inverter pair on a single, non-series-stacked machine [99], and was later applied to a genuine two-cell series-stacked (2L-SSC) SPB inverter [58]; as with interleaved GaN IMMDs, the achievable reduction is ultimately limited by layout parasitics and local dc-link capacitor design [40].

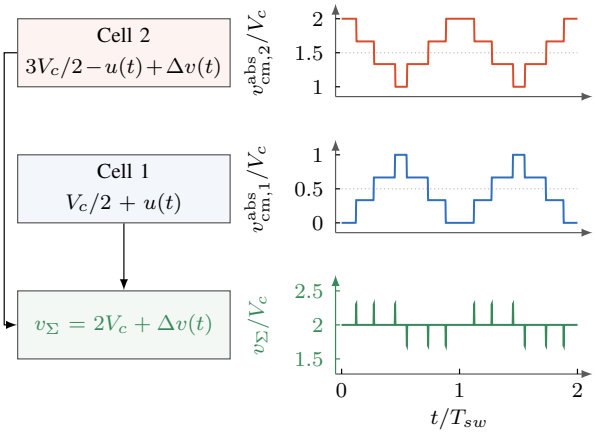


FIGURE 11. Nonideal two-cell common-mode cancellation over two switching periods. Small transition misalignment produces narrow residuals in v_Σ.

This cancellation assumes perfectly matched, simultaneous switching transitions between the two cells: a propagation-delay mismatch (e.g., from PCB layout or driver differences) reintroduces a residual CM spike, of duration equal to the delay, at every misaligned transition, and unequal power sharing during dc-link balancing similarly degrades the cancellation, both shown quantitatively in [58], [99]. Fig. 11 illustrates the reference levels as an example specific to $N = 2$. For fixed lumped winding-to-frame capacitances C_{kg}, the frame excitation is approximately

$$i_{\mathrm{frame}} = \sum_k C_{kg} \frac{d}{dt}(v_{\mathrm{cm},k}^{\mathrm{abs}} - v_{\mathrm{frame}}). \tag{17}$$

A constant stack offset contributes insulation stress but no steady capacitive current.

C. Physical Consequences and Mitigation

CMV couples capacitively to the rotor and can cause bearing discharges when the lubricant film breaks down [100]; non-intrusive current measurements support diagnosis [101]. In SPB, the winding groups have different absolute CM references (14)–(15). Their combined contribution to shaft voltage and bearing current, including possible resonances, requires stack-level assessment.

Fast winding-terminal transitions also stress turn-to-turn and phase-to-frame insulation. Short rise times and overshoot associated with WBG switching can accelerate insulation aging and partial discharge [102]–[104]. Insulation diagnostics [105] and suitable material systems [106] complement coordination of the local switching stress and the stack-position-dependent offset.

Candidate mitigation measures include active EMI filtering [107], shaft-voltage cancellation [108], and CMV-aware zero-sequence injection or redundant-vector selection [89], [90]. However, carrier phasing selected for ripple reduction (8) does not necessarily minimize aggregate CM excitation and cannot alter the fixed stack offset $V_{-,k}$. SPB modulation and insulation design should therefore evaluate ripple, CM excitation, and insulation stress jointly across the full stack.

VII. DC-Link Dynamics, Stability, and Voltage Balancing

The common dc current and independently controlled winding-group powers couple the local capacitor energies, so equal cell voltages are not inherently maintained [45], [46]. Related research likewise shows that distributed current control alone does not ensure stable voltage sharing [92].

For cell k, the power balance is

$$C_k V_{\mathrm{dc},k} \frac{dV_{\mathrm{dc},k}}{dt} = V_{\mathrm{dc},k} i_{\mathrm{dc}} - p_{\mathrm{ac},k} - p_{\mathrm{loss},k}. \tag{18}$$

Because the series current is common, dc input power depends on local voltage, whereas the winding-group controller determines ac output power. Defining the deviation from the mean cell voltage as

$$\widetilde{V}_k = V_{\mathrm{dc},k} - \frac{1}{N}\sum_{j=1}^{N} V_{\mathrm{dc},j}. \tag{19}$$

a simplified differential-mode model around a balanced operating point is

$$C V_0 \frac{d\widetilde{V}_k}{dt} \approx I_{\mathrm{dc},0}\widetilde{V}_k - \Delta P_k. \tag{20}$$

where C and V_0 are the common cell capacitance and equilibrium voltage, $I_{\mathrm{dc},0}$ is the equilibrium dc current, and ΔP_k is the differential output-power perturbation. During motoring, $I_{\mathrm{dc},0} > 0$, a positive voltage perturbation increases cell input power and reinforces the deviation if output power remains unchanged. During regeneration, this feedback reverses. This interpretation agrees with SPB stability analyses [45], [46] and related series-connected multiphase-drive studies [109].

The constant-power-load model considered in [46] comprises

$$\begin{aligned} L_b \frac{di_{\mathrm{dc}}}{dt} &= E_b - R_b i_{\mathrm{dc}} - \sum_k V_{\mathrm{dc},k}, \\ C_k \frac{dV_{\mathrm{dc},k}}{dt} &= i_{\mathrm{dc}} - \frac{P_k}{V_{\mathrm{dc},k}}. \end{aligned} \tag{21}$$

Here, E_b is the dc-source voltage, and L_b and R_b are the lumped series inductance and resistance of the shared source-to-stack supply path, including source and interconnection contributions; P_k is the cell's constant power demand. Under this model's assumptions, linearization about the open-loop motoring equilibrium for $N > 1$ yields $N - 1$ strictly positive real eigenvalues, establishing instability irrespective of parameter selection.

Fig. 12 illustrates the balancing principle using the small-signal model in (20), consistent with [46]. At equal constant output power, a lower-voltage cell draws more converter input current ($P_k/V_{\mathrm{dc},k}$) and its capacitor discharges further; a higher-voltage cell exhibits the opposite response. Thus, a small mismatch grows during motoring unless differential power is adjusted. The law $\Delta P_k = K_p \widetilde{V}_k$, motivated by controller alternative I in [46], increases output power in

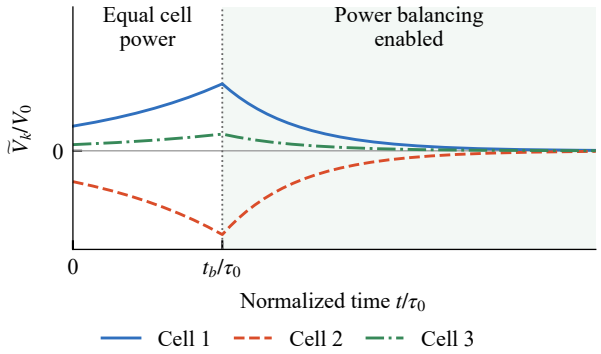


FIGURE 12. Conceptual analytical illustration of voltage balancing for three cells during motoring, based on (20) and [46]. Small deviations from the mean grow under equal constant cell powers. Enabling differential power balancing at t_b makes them decay toward zero while $\sum_k \Delta P_k = 0$. Time is normalized by $\tau_0 = CV_0/I_{\mathrm{dc},0}$.

higher-voltage cells and decreases it in lower-voltage cells. For $K_p > I_{\mathrm{dc},0}$, (20) predicts exponential decay of the deviations. This local result assumes identical capacitances and ideal power-command tracking; it does not imply instability under every converter–machine control strategy.

The winding groups provide control authority over internal capacitor energy through differential power commands:

$$P_k^\star = \frac{P^\star}{N} + \Delta P_k, \qquad \sum_{k=1}^{N} \Delta P_k = 0, \tag{22}$$

This allocation redistributes cell power without changing the total machine-power reference. Available balancing authority depends on phase-current margin, winding voltage, torque-ripple limits, and thermal state.

Table 4 summarizes three principal approaches. Centralized control explicitly regulates voltage sharing through differential commands [45], [46]. Common-duty-ratio (CDR) control instead exploits the coupled converter–machine dynamics; stable sharing has been demonstrated for the investigated induction-machine system [53], with practical performance sensitive to timing mismatch [56]. Distributed multi-agent control extends local coordination to voltage balancing, power sharing, and reconfiguration [54], [55], [57]. Related modular-drive work uses local speed-droop laws to coordinate module power sharing [110].

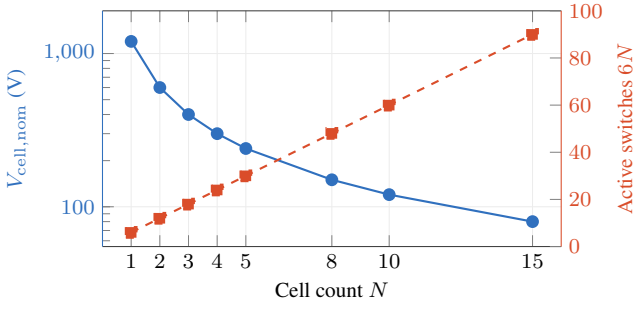
For automotive applications, balancing must remain effective during torque transients, field weakening, parameter mismatch, and module reconfiguration. Sensor offsets, communication latency, propagation delays, and differential power limits determine the usable cell-voltage margin and therefore influence semiconductor voltage-rating selection. Dc-link stability must consequently be assessed across the coupled converter, machine, and control system.

VIII. Semiconductor Technology and Cell-Count Scaling

Increasing the SPB cell count N enables lower-voltage semiconductor devices, but voltage-rating selection must account

TABLE 4. Principal SPB dc-link balancing approaches and their main considerations.

Approach	Mechanism	Key consideration
Centralized	Differential current/power commands [45], [46]	Direct authority; global sensing and coordination
CDR	Common modulation, natural coupled dynamics [53], [56]	Low overhead; symmetry- and timing-sensitive
Distributed	Local agents, neighbor communication [54], [55], [57]	Scalable; requires communication and synchronization



N	1	2	3	4	5	8	10	15
V_{cell} (V)	1200	600	400	300	240	150	120	80
$6N$	6	12	18	24	30	48	60	90

FIGURE 13. Cell-count scaling for a 1200-V dc link: nominal cell voltage $V_{\mathrm{cell,nom}} = V_{\mathrm{dc}}/N$ and active switch positions $6N$.

for maximum blocking stress:

$$V_{\mathrm{rated}} \geq k_{\mathrm{rel}} \left(\frac{V_{\mathrm{dc,max}}}{N} + \Delta V_{\mathrm{imb}}^{\mathrm{max}} + \Delta V_{\mathrm{sw}}^{\mathrm{max}} \right), \tag{23}$$

where $\Delta V_{\mathrm{imb}}^{\mathrm{max}}$ is the maximum cell-voltage imbalance, $\Delta V_{\mathrm{sw}}^{\mathrm{max}}$ is the commutation overshoot, and k_{rel} is the selected voltage-rating margin factor. Cell-count selection therefore depends on discrete commercial voltage classes and their usable operating margins.

SiC is well suited to high-voltage traction, while voltage partitioning can make lower-voltage GaN devices attractive through favorable conduction and switching characteristics [25], [36], [58]. For the illustrative 1200-V dc link in Fig. 13, two cells operate at a nominal 600 V each, leaving only 50 V of headroom for 650-V devices. This allocation is infeasible if bus variation, imbalance, switching overshoot, and the selected design margin require more than this headroom. Three cells reduce this voltage to 400 V, providing greater allowance for imbalance and switching transients. Further increases in N enable lower-voltage device classes but increase the number of active switch positions as $6N$. Final device selection must satisfy (23).

A. Commercial-Device Screening

Lower voltage ratings do not guarantee superior converter performance. Candidate devices must be assessed using con-

duction and switching characteristics together with current capability, thermal limits, and packaging constraints. Figures of merit such as $R_{\mathrm{DS(on)}}C_{\mathrm{oss}}$ and $R_{\mathrm{DS(on)}}Q_g$ provide initial screening, but detailed comparison requires $R_{\mathrm{DS(on)}}(T)$ for conduction loss and die-area requirements; $E_{\mathrm{oss}}(V)$ for stored output-capacitance energy and its contribution to switching loss; $E_{\mathrm{on/off}}(V, I, T)$ for switching-energy characterization; Q_g and Q_{gd} for gate-drive requirements and switching behavior; I_{cont} and R_θ for current derating, parallelization, and cooling; and package and power-loop inductance L_σ for commutation overshoot.

Machine phase current does not generally decrease as $1/N$. For a derated device current capability $I_{\mathrm{dev,der}}$, the required parallel-device count and total active-device count can be estimated as

$$n_p = \left\lceil \frac{I_{\mathrm{sw,max}}}{I_{\mathrm{dev,der}}} \right\rceil, \qquad N_{\mathrm{sw,tot}} = 6Nn_p. \tag{24}$$

Consequently, favorable device-level figures of merit may coexist with substantial die area, package count, and gate-drive and thermal-interface requirements.

Device selection and layout are further coupled through commutation overshoot:

$$\Delta V_{\mathrm{sw}} \approx L_\sigma \frac{di}{dt}, \tag{25}$$

Package and interconnection inductance can consume much of the voltage margin gained through partitioning. High-current GaN modules illustrate the importance of low-inductance packaging and integrated gate driving at traction-relevant currents [111], [112]. High switching speeds therefore require careful commutation-loop design, local decoupling, and gate-drive layout, particularly for GaN [112]. Dynamic $R_{\mathrm{DS(on)}}$, reverse conduction, gate reliability, short-circuit capability, and thermal performance must also be considered [113], [114]. Room-temperature static resistance alone is therefore insufficient for cell-count selection.

B. Loss Scaling With Cell Count

For an individual device, conduction and switching losses can be approximated as

$$P_{\mathrm{cond}} \approx I^2_{\mathrm{rms,sw}} R_{\mathrm{DS(on)}}(T_j), \tag{26}$$

and

$$P_{\mathrm{sw}} \approx f_{\mathrm{sw}} \left[E_{\mathrm{on}}(V, I, T_j) + E_{\mathrm{off}}(V, I, T_j) \right]. \tag{27}$$

Including dead-time-related losses and gate-drive power gives

$$P_{\mathrm{semi}}(N) = \sum_{k=1}^{N} \sum_{j=1}^{6n_p(N)} \left(P_{\mathrm{cond},k,j} + P_{\mathrm{sw},k,j} + P_{\mathrm{dt},k,j} \right) + P_{\mathrm{gate}}(N), \tag{28}$$

where $P_{\mathrm{dt},k,j}$ represents dead-time-related device loss and P_{gate} is the aggregate gate-drive power.

Total loss does not scale linearly with N: additional cells increase device count, while transitions between voltage classes change on-state resistance, switching energy, thermal impedance, and parallelization requirements. Low-cell-count designs may favor high-voltage SiC, whereas lower-voltage GaN becomes attractive when its device, packaging, or passive-component benefits offset the additional modules.

IX. Fault Tolerance and Reconfiguration

SPB modularity offers potential graceful degradation because the converter and machine are divided into repeated converter–winding modules. However, fault tolerance is not automatic: because the dc ports are connected in series, a failed cell can disturb or interrupt the complete dc current path, a structural challenge demonstrated for SPB and for other series-connected modular converters alike [115]–[117]. Early SPB work demonstrated post-fault operation under semiconductor faults, establishing that repeated hardware must be accompanied by appropriate safe states and control action [115]. At the machine level, a submodule short-circuit fault in a modular converter feeding fractional-slot concentrated-winding PM machines was shown to drive a fault current above rated current because of the strong NdFeB rotor magnetization, risking machine overheating; reducing rotor magnetization limits this fault current but also narrows the constant-power-speed range, a design tradeoff specific to fault-tolerant FSCW PM machines [116]. In the SPB machine designs of that study, post-fault short-circuit current with NdFeB magnets exceeds the healthy-phase current, whereas with ferrite magnets it stays below it, at the cost of becoming visibly non-sinusoidal from magnetic coupling with the adjacent healthy windings – a concrete illustration that PM material selection is itself a fault-tolerance design variable, not only a torque-density one. Fault handling in other series-connected modular converters further underscores that mitigation requires the control system to be explicitly designed for module isolation [117].

The machine provides a second layer of redundancy. Healthy winding groups can potentially redistribute current after phase or converter faults, as demonstrated in recent multiphase and modular-drive studies [63], [118], [119]. In SPB, however, this machine-side redundancy is useful only if the failed converter cell can be isolated or bypassed without breaking the series dc path.

Building on the reported online reconfiguration capability [57], after a faulted module is isolated the remaining active set $\mathcal{A}$ must satisfy

$$P^\star = \sum_{k \in \mathcal{A}} P^\star_k, \tag{29}$$

with torque derating if current, voltage, or thermal limits prevent full redistribution. Reconfiguration can therefore provide both fault isolation and an operating-range degree of freedom because the number of active cells changes the voltage available to the remaining winding groups. Bypassing a cell raises the steady voltage of each surviving cell when the total bus voltage remains fixed. With $M = |\mathcal{A}|$

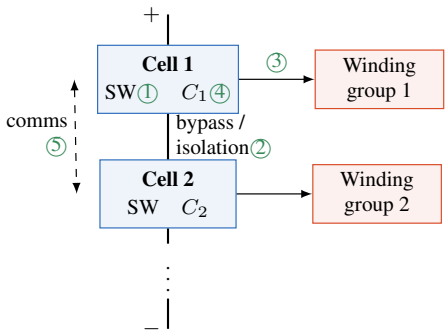


FIGURE 14. **Simplified SPB converter–machine structure with fault locations ①–⑤ annotated, corresponding in order to the rows of Table 5.**

TABLE 5. **SPB Fault Locations and Required System Response**

Fault	SPB consequence	Required response
Switch / bridge leg	Cell cannot synthesize commanded voltage	Fast safe state and protection [115]
Converter module	Series dc path can be interrupted	Bypass/isolation and power redistribution [57]
Winding / phase	Reduced and asymmetric torque capability	Multiphase post-fault current control [63], [119]
Capacitor / sensor*	Loss of voltage-sharing information or margin	Detection, derating, isolation or shutdown
Communication / controller	Loss of distributed coordination	Local fallback and deterministic safe state [120]

*The SPB studies surveyed here do not establish capacitor or sensor fault coverage; this row reflects the authors' system-level inference.

active cells, a necessary blocking-voltage condition is

$$\frac{V_{\mathrm{dc,max}}}{M} + \Delta V_{\mathrm{imb}} + \Delta V_{\mathrm{sw}} \le V_{\mathrm{allow}}, \tag{30}$$

where V_{allow} is the allowable device voltage after the chosen design margin, and the two increments cover cell imbalance and switching overshoot. Capacitor and insulation limits must also hold. Current or torque derating cannot by itself correct excessive steady blocking voltage. If this condition fails, operation requires a lower bus voltage, a suitable spare-cell arrangement, or shutdown. Fig. 15 separates voltage feasibility from post-fault torque capability. Fig. 14 maps these fault locations onto the SPB structure.

Reliability remains ambiguous because increasing cell count provides more potential degraded states but also more components that can fail. System or mission reliability is therefore more meaningful than component count alone. Markov-chain analysis has been applied to related fault-tolerant IMMDs to distinguish healthy, degraded, and failed states [121], while integrated distributed traction drives provide additional evidence that fault tolerance must include both the machine and the local converter architecture [71], [72].

The SPB literature therefore demonstrates the feasibility of post-fault operation and online reconfiguration, but not yet automotive-level fault coverage. Detection and isolation time, short-circuit energy, capacitor and sensor faults, communication loss, post-fault thermal loading, diagnostic coverage, and deterministic functional-safety transitions remain important validation topics.

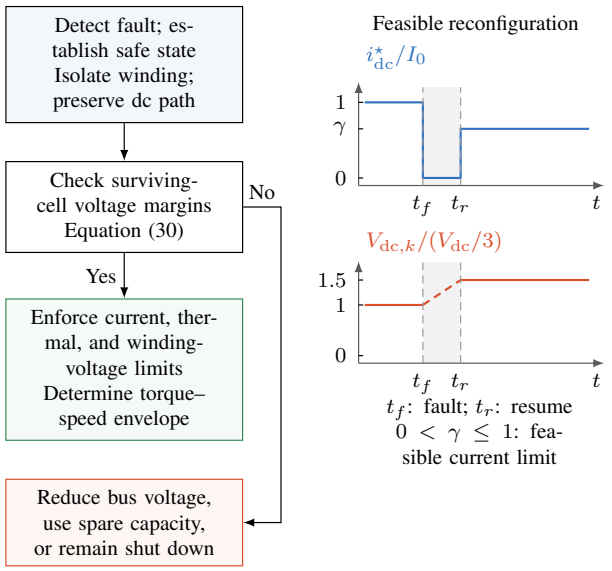


FIGURE 15. **Post-fault feasibility assessment with illustrative waveforms for the feasible branch. The dc-current command is temporarily set to zero and restored to γI_0, subject to operating limits. For a fixed bus, changing from three to two active cells raises the balanced surviving-cell voltage from 1 to 1.5 p.u. (400 to 600 V for a 1200-V bus).**

TABLE 6. **300-kW SPB design inputs and preliminary semiconductor-loss estimate**

Output power / dc-bus voltage	300 kW / 1200 V
Cells / nominal cell voltage	16 / 75 V
Devices per switch / drivers per leg	4 / 2
Total devices / half-bridge drivers	384 / 96
Phase current / switching frequency	227.7 A rms / 50 kHz
MLCCs per phase leg	$20 \times 22 + 40 \times 1$ μF
Phase-leg capacitance, nominal / effective	480 / 134.2 μF
PCB footprint per phase leg	76.25 cm^2
Semiconductor-only efficiency / loss	99.71% / 0.865 kW
Ideal shared-bus dc-link ripple	2.85% (target 3%)
Screening junction temperature	68.8 °C
Updated four-parallel leg: extracted layout	
Board size / layer count	89.5 × 85.2 mm / 4
Power loop: load-test layout	0.409 nH

Loss component	Leg (W)	Stack (W)
Conduction	12.52	601.0
Switching	4.96	238.3
Dead time	0.46	22.3
Gate drive	0.07	3.4
Total	18.02	864.9

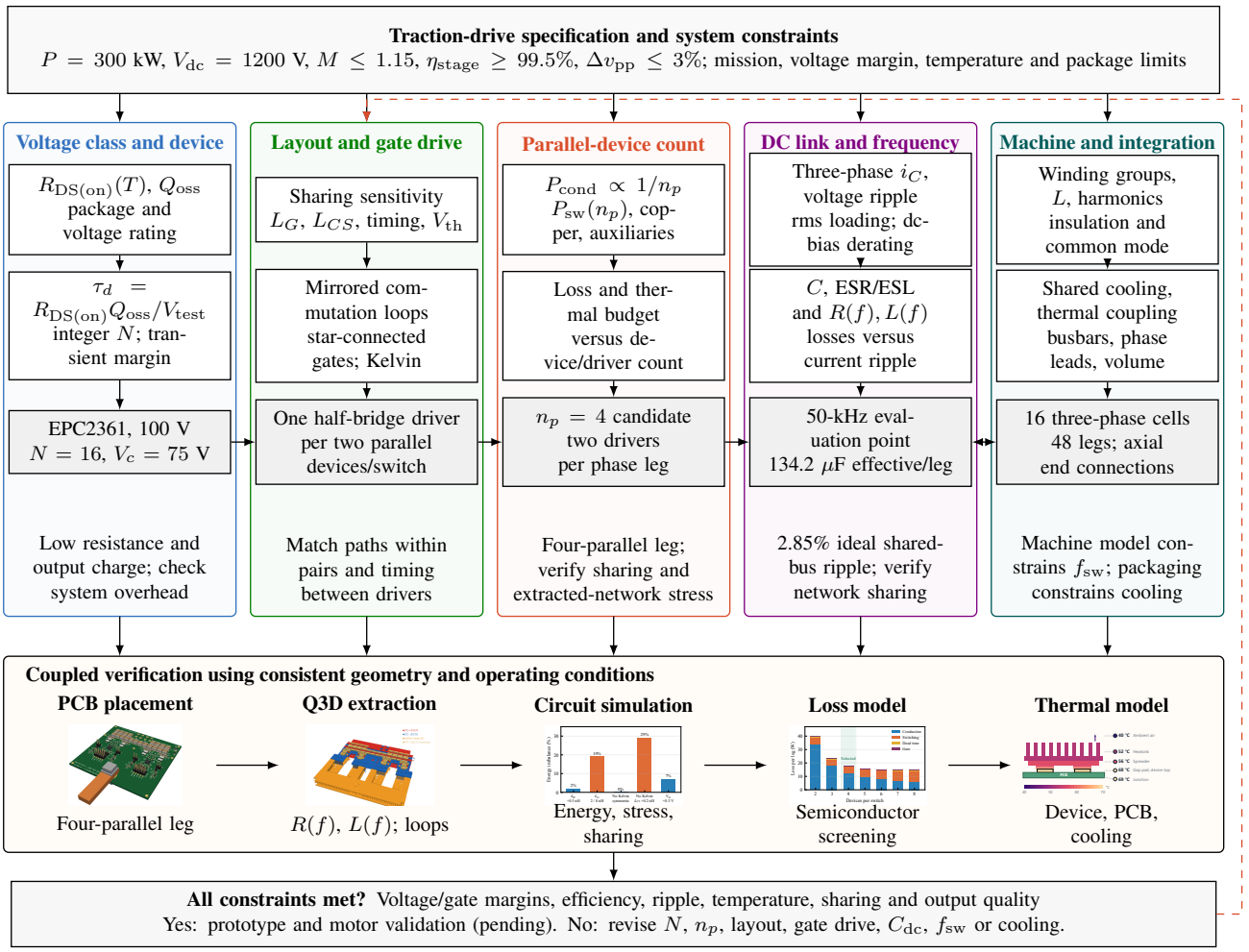


FIGURE 16. **Iterative SPB system-design procedure with coupled verification. Machine requirements feed back into switching-frequency, insulation, and thermal design. Updated four-parallel building-block models support preliminary screening; shared cooling and interconnections require assembly-level and experimental validation.**

X. Illustrative Device-to-System Design Example

Fig. 16 synthesizes the preceding review into an iterative device-to-system design procedure. Electrical specifications and the operating mission define the constraints; device voltage class, cell count, layout, parallel count, capacitance, and switching frequency are then evaluated together with the machine and cooling arrangement. The verification loop revisits these choices until electrical, thermal, and packaging requirements are met. It provides a framework for system optimization, rather than demonstrating a global optimum.

The example in Table 6 uses a 300-kW, 1200-V SPB. For balanced sinusoidal operation, $I_{ph,rms} = 2\sqrt{2}P_{out}/(3MV_{dc}\cos\varphi)$; $M = 1.15$ and a power factor of 0.9 give 227.7 A rms. The EPC2361 is selected as an illustrative 100-V candidate [122]. Sixteen cells operate at 75 V nominal; maximum-bus operation, balancing error, and overshoot must still satisfy (23).

Four devices per switch use matched Kelvin gate paths and two half-bridge drivers per leg. At 50 kHz, the updated screening model includes temperature-dependent conduction, switching, dead-time reverse conduction, and gate-drive losses. With a 0.5-nH commutation loop, 40-°C ambient, and a 0.65-K/W leg-to-air thermal resistance, it gives 18.02 W per leg and a junction temperature of 68.8 °C. Four devices are the smallest evaluated parallel count meeting the initial temperature and voltage constraints; this screening does not establish a system optimum.

The revised bank uses twenty 22-μF MLCCs and forty 1-μF bypass capacitors per leg. After dc-bias derating at 75 V, their effective capacitance is 134.2 μF per leg, or 402.6 μF per three-phase cell. The balanced three-phase switching-charge model predicts 2.85% ripple at 50 kHz under ideal shared-bus conditions, against a 3% target. Network-dependent current sharing, damping, and capacitor losses require further verification.

The estimated efficiency of 99.71% includes semiconductor and gate-drive losses only; PCB and busbar losses, capacitor ESR, external supplies, control electronics, cooling auxiliaries, and machine losses are excluded. Extracted layout parameters are listed in Table 6. Assembly-level and

experimental validation remain necessary. The air-cooled thermal result does not validate the shared-cold-plate motor-integration concept.

XI. Research Gaps and Future Directions

SPB competitiveness depends on joint battery, converter, and machine design. Voltage partitioning, winding constraints, modulation, balancing, common-mode voltage, and fault tolerance must be evaluated together.

The optimum cell count N remains the central unresolved design question: the voltage-versus-component-count tradeoff shifts the optimum as SiC, GaN, packaging, and cooling evolve [3], [25]. A common-platform comparison of a few-cell 650-V GaN SPB against a many-cell low-voltage GaN SPB, using real commercial devices, would be valuable; the GaN comparison in [58] is a useful starting point but should be extended to automotive power with current parallelization, machine voltage/speed constraints, dc-link capacitance, cooling, and mission energy.

Converter, machine, and thermal models, among others, should be co-optimized rather than designed sequentially. Existing phase-count studies show machine design changes with SPB cell number [34]; the missing step is a complete automotive co-design using vehicle-level energy, volume, mass, cost, and reliability as a joint objective.

Reliability is part of that objective: future studies should combine semiconductor, capacitor, sensor, gate-driver, communication, and winding faults [57], [115] in one state-based, Markov-style framework as used for related fault-tolerant IMMDs [121], keeping *component failure*, *module isolation*, and *loss of vehicle function* distinct. An N-cell drive may survive with $N-1$ active modules, but its torque–speed envelope, module thermal loading, and dc-voltage margin all change, so reliability data should pair with a post-fault capability model – not a series reliability block diagram – to compare availability, limp-home power, and mission completion probability against efficiency.

Cost and sustainability assessment should use the same N-dependent framing: increasing N can reduce per-switch die cost but tends to increase isolated gate supplies, sensing channels, and assembly operations, among others, almost linearly, while a repeated converter–winding module can gain from common tooling and automated assembly – a balance likely to differ between high-volume passenger vehicles and lower-volume heavy-duty systems. The same accounting should separate production, operation, maintenance, and end-of-life impact over one vehicle mission and service life, since modular repair and longer service life can offset the added production impact of extra components.

Overall, SPB research must move from demonstrating that the architecture operates toward demonstrating where it is better: whether it gives a measurable vehicle-level advantage over mature two- and three-level architectures once machine design, thermal management, EMC, reliability, manufacturing, and cost are included.

XII. Conclusion

This paper reviewed SPB converters – series-stacked two-level bridges each feeding an electrically separated winding group – as a candidate architecture for future high-voltage traction drives, partitioning the dc-link voltage while introducing converter–machine modularity, distributed control, and fault-management opportunities.

The review also shows that the semiconductor voltage advantage cannot be separated from the complete traction system. Increasing cell count adds converter, control, and insulation overhead, so SPB does not eliminate the complexity of high-voltage conversion but redistributes it between semiconductor technology, machine architecture, and control. WBG development makes this trade increasingly relevant, particularly where SPB can move a high-voltage traction system toward lower-voltage GaN devices; whether that approach is advantageous depends on whether the device-level gains compensate for the additional modular hardware and machine requirements.

The remaining priority is integrated co-design across the battery/dc-link, semiconductor/converter, and machine sides, evaluated jointly against vehicle-level energy, cost, and reliability targets rather than compared through separate, narrower studies. The proposed design framework makes the evaluation boundary explicit. SPB is a modular traction-drive architecture that exchanges semiconductor blocking-voltage stress for converter–machine modularity. Its competitiveness depends on whether that exchange produces a measurable vehicle-level benefit.

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